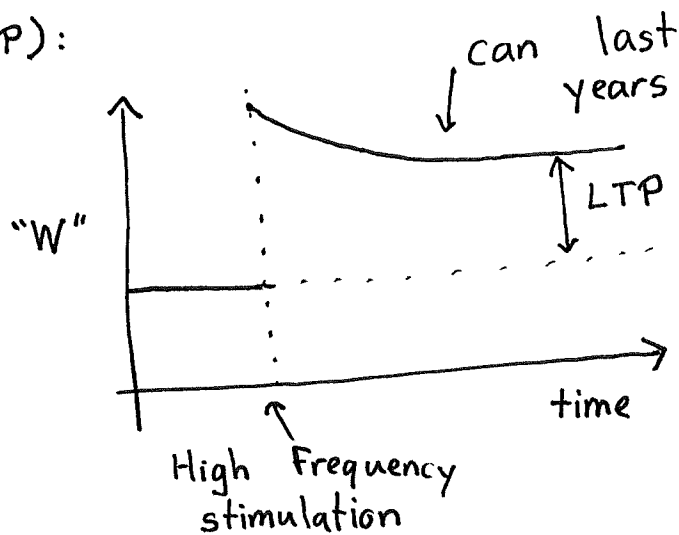
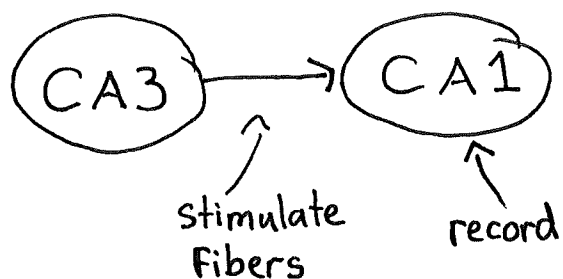


Synaptic plasticity:

In 1949, Donald Hebb hypothesized:

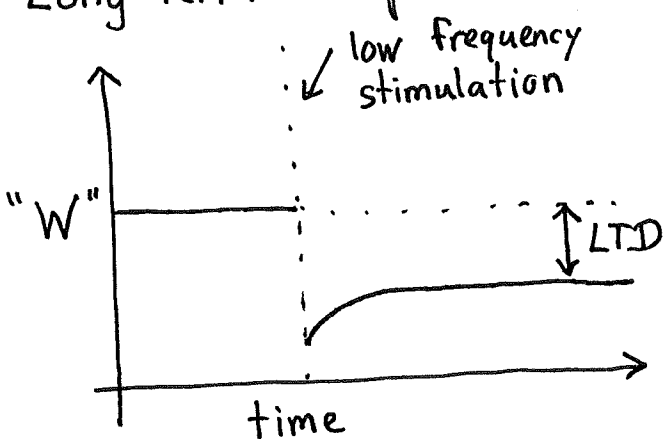
Hebb's Law: "Neurons that fire together wire together."

Long-term potentiation (LTP):

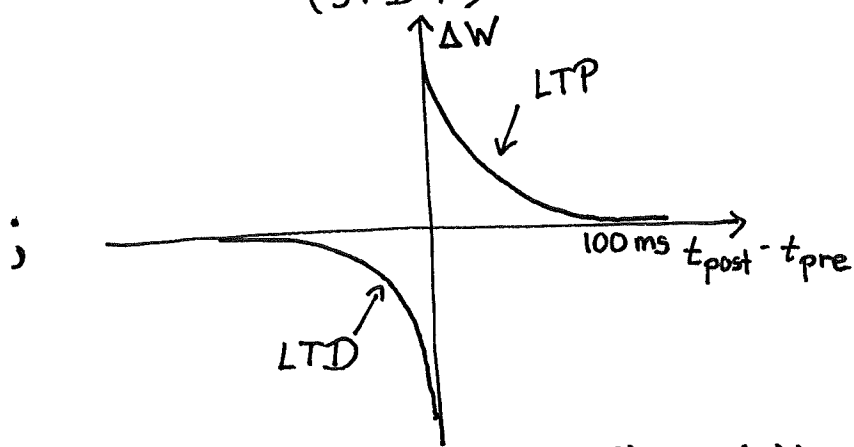


Generalized Hebb's Law: "Neurons that don't, don't."

Long-term depression (LTP):



Spike-timing dependent plasticity (STDP)



$$W_{ij} \leftarrow W_{ij} + f(t^{(i)} - t^{(j)})$$

Mechanism:

Weight changes due to AMPA insertion or deletion.
Insertion ^(High Ca) vs. deletion ^(Low Ca) due to post-synaptic calcium level.
Post-synaptic calcium high when NMDA channels active.
NMDA channels active when pre precedes post, because pre releases Mg block.

Hebbian learning in firing rate models:

Basic Hebb rule:

$$\tau_w \frac{dW_{ij}}{dt} = h_i h_j$$

Can occur in feedforward or recurrent synapses (i.e., u or W in previous notation). No depression if $h_i, h_j \geq 0$.

Generalized Hebb rules:

$$(i) \tau_w \frac{dW_{ij}}{dt} = (h_i - \langle h_i \rangle) h_j$$

$$(ii) \tau_w \frac{dW_{ij}}{dt} = h_i (h_j - \langle h_j \rangle)$$

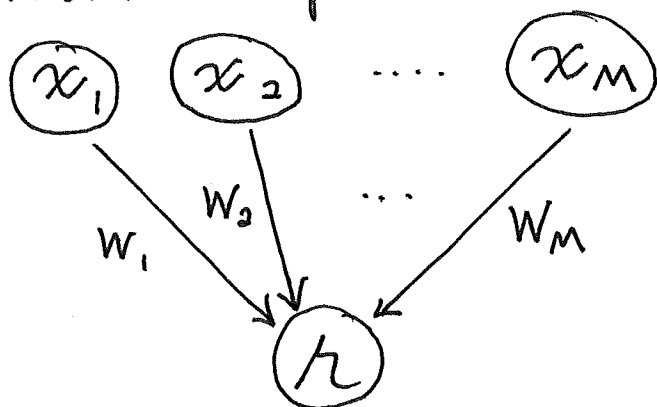
$$(iii) \tau_w \frac{dW_{ij}}{dt} = (h_i - \langle h_i \rangle)(h_j - \langle h_j \rangle)$$

The expected neural activity, $\langle h_i \rangle$, is ~~over~~ an average over whatever randomness is in the inputs driving activity on thus plasticity. All three are called "covariance rules," as

$$\tau_w \left\langle \frac{dW_{ij}}{dt} \right\rangle = C_{ij}^{(h)} ; \langle \cdot \rangle : \text{same average as above}$$

where $C_{ij}^{(h)}$ is the i, j element of the covariance matrix of neural firing rates. A major problem with this rule is that it is unstable.

Feedforward example:



General:

$$h = \mathcal{F}\left(\sum_{j=1}^M W_j x_j\right)$$

Threshold-linear / Linear

$$h = \sum_{j=1}^M W_j x_j$$

$$= W^T x$$

In terms of x , learning rule (ii) is

$$\tau_w \frac{dW_j}{dt} = \sum_{k=1}^M W_k x_k (x_j - \langle x_j \rangle)$$

The expected weight dynamics are thus

$$\tau_w \left\langle \frac{dW_j}{dt} \right\rangle = \sum_{k=1}^M \langle W_k x_k (x_j - \langle x_j \rangle) \rangle$$

$$\tau_w \frac{d\bar{W}_j}{dt} = \sum_{k=1}^M \bar{W}_k C_{kj}^{(x)} = (\bar{W}^T C^{(x)})_j$$

where $\bar{W}$ is the average weight vector we expect by averaging over randomness in the inputs. We will soon see that the weights never stabilize due to a positive feedback loop.

large weights $\rightarrow$ large change $\rightarrow$ larger weights $\rightarrow$ larger change $\rightarrow$

Aside on linear algebra:

Consider a matrix, M , of size $N \times N$. An $N \times 1$ vector, v , is an eigenvector of M with eigenvalue λ if

$$Mv = \lambda v$$

where λ is a number. In words, M acts on its eigenvectors by scaling alone. Note, av is also eigenvector.

Similarly, v is a left eigenvector of M with eigenvalue λ if

$$v^T M = \lambda v^T$$

Suppose that M is symmetric

$$M_{ij} = M_{ji}, \quad M^T = M$$

Then,

$$Mv = \lambda v \Rightarrow v^T M^T = \lambda v^T \Rightarrow v^T M = \lambda v^T$$

For symmetric matrices, each eigenvector is also a left eigenvector.

Suppose M is a covariance matrix

$$M_{ij} = \langle (x_i - \langle x_i \rangle)(x_j - \langle x_j \rangle) \rangle, \text{ NOTE: } M \text{ is symmetric}$$

Then its eigenvalues are non-negative.

$$Mv = \lambda v \Rightarrow v^T M v = \lambda v^T v = \lambda \|v\|^2$$

On the other hand,

$$\begin{aligned} v^T M v &= \sum_i \sum_j v_i \langle (x_i - \langle x_i \rangle)(x_j - \langle x_j \rangle) \rangle v_j \\ &= \langle y^2 \rangle \geq 0 \end{aligned}$$

where $y = \sum_i v_i (x_i - \langle x_i \rangle)$. Thus $\lambda = \langle y^2 \rangle / \|v\|^2 \geq 0$.

Back to Hebbian plasticity:

We have

$$\tau_w \frac{d\bar{w}}{dt} = \bar{w}^T C(x)$$

Suppose

$$w(0) = \bar{w}(0) = \tilde{w}$$

is an eigenvector of $C(x)$. Then $\tilde{w}^T C(x) = \lambda \tilde{w}^T$, and

$\frac{d\bar{w}}{dt}$ is in the direction of $\tilde{w}$. Thus $\bar{w}$ stays aligned to $\tilde{w}$. Hebbian plasticity changes amplitude of $\bar{w}$, but not the direction. $\bar{w}$ is always an eigenvector

$$\tau_w \frac{d\bar{w}}{dt} = \lambda \bar{w}^T$$

$$\tau_w \frac{d\bar{w}_i}{dt} = \lambda \bar{w}_i$$

$$\bar{w}_i = \tilde{w}_i \underbrace{e^{t\lambda/\tau_w}}$$

$\lambda \geq 0$. Exponential growth of amplitude

Routes to stabilization:

Synaptic scaling: Rescale weights during dynamics to ensure $\|\bar{W}\|^2 = \|\tilde{W}\|^2$. For most initializations, this leads $\bar{W}$ to align with ~~eigenvector~~^{vector} of $C^{(\infty)}$ with largest eigenvalue. PCA!

Explicit weight dependence:

$$\tau_w \frac{dW_{ij}}{dt} = \underbrace{\kappa_i \kappa_j}_{\text{Hebbian}} - \underbrace{\kappa_i^2 W_{ij}}_{\text{Weight decay}} \quad (\text{Oja's rule})$$

Also does PCA in feedforward case. Solution has norm 1.

Dynamic threshold:

$$\tau_w \frac{dW_{ij}}{dt} = \kappa_i \kappa_j (\kappa_i - \theta) \quad (\text{BCM rule})$$

$$\tau_\theta \frac{d\theta}{dt} = \kappa_i^2 - \theta$$

Requires both presynaptic and postsynaptic activity. Sign of change set by magnitude of postsynaptic activity.

Beyond Hebbian Plasticity:

Hebbian learning is "unsupervised." It can find interesting patterns (e.g.: largest eigenvectors of input's covariance matrix; PCA), but animals often need to do more than recognize patterns. Depends on neural activity pre & post

"Supervised" learning: Here inputs are paired with target outputs. Learning rules depend on error signals like

$$e = (y - \hat{y}(x, w)),$$

where y is target output and $\hat{y}$ is current output given the input (x) and current weights (w). ^{For example} ~~The cerebellum~~, the cerebellum is hypothesized to implement supervised learning with error signals from inferior olive.

Reward learning: Here learning rules serve to associate stimuli and/or actions with rewarding outcomes. They depend on reward signals, R , and/or expected reward signals, which are thought to be represented by dopamine ~~neurons~~ neuron activity.

"Three-Factor" rules: idea that plasticity depends on: (i) pre-synaptic activity; (ii) post-synaptic activity; (iii) a "third factor" that ~~gates~~ indicates when learning should occur (e.g., error signal, reward, reward prediction error).